Rearrangements of the Alternating Harmonic Series

Colin McConnell

Source: Motivated by S. Abbott, *Understanding Analysis*, Chapter 2

Setup and Statement

Consider the alternating harmonic series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots .$$

We separate its terms into two sequences. The positive terms are

$$p_k = \frac{1}{2k-1} \qquad (k = 1, 2, 3, \dots),$$

so that $p_1 = 1$, $p_2 = \frac{1}{3}$, $p_3 = \frac{1}{5}$, and so on. The negative terms are $-q_1, -q_2, -q_3, \dots$, where

$$q_k = \frac{1}{2k} \qquad (k = 1, 2, 3, \dots),$$

so that $q_1 = \frac{1}{2}$, $q_2 = \frac{1}{4}$, $q_3 = \frac{1}{6}$, and so on.

Definition. A *rearrangement* of the alternating harmonic series is a series $b_1 + b_2 + b_3 + \cdots$ in which each term of the original series appears exactly once.

Theorem. *For every real number L, there exists a rearrangement of the alternating harmonic series that converges to L.*

The proof proceeds in three stages. First, we establish two properties of the sequences (p_k) and (q_k). Second, we define a rule that determines the order of the terms. Third, we verify that this rule produces a rearrangement and that the rearrangement converges to L.

Two Properties of the Sequences

Fact (1)**.** *$p_k \to 0$ and $q_k \to 0$.*

Proof. For every $k \in \mathbb{N}$, we have $0 < q_k < p_k = \frac{1}{2k-1} \leq \frac{1}{k}$. Since $\frac{1}{k} \to 0$, both sequences converge to 0. □

Fact (2)**.** *For every $j \in \mathbb{N}$, the partial sums*

$$p_j + p_{j+1} + \cdots + p_n \qquad \text{and} \qquad q_j + q_{j+1} + \cdots + q_n$$

are unbounded above as $n \to \infty$.

Proof. We use the fact that the harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$ diverges, so that its partial sums $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$ are unbounded above.

Since $q_k = \frac{1}{2k}$, we have

$$q_1 + q_2 + \cdots + q_n = \frac{1}{2} H_n,$$

which is unbounded above. Moreover, $p_k > q_k$ for every k, so

$$p_1 + p_2 + \cdots + p_n > q_1 + q_2 + \cdots + q_n,$$

and therefore the partial sums of the p_k are also unbounded above. This proves the case $j = 1$.

Now let $j > 1$. Then

$$p_j + p_{j+1} + \cdots + p_n = (p_1 + \cdots + p_n) - (p_1 + \cdots + p_{j-1}).$$

The second quantity on the right is a fixed number that does not depend on n, and subtracting a fixed number from an unbounded sequence yields an unbounded sequence. The same argument applies to the q_k. □

Construction of the Rearrangement

Fix $L \in \mathbb{R}$. We define the terms $b_1, b_2, b_3, \ldots$ one at a time. Let $s_n = b_1 + b_2 + \cdots + b_n$ denote the n-th partial sum, and set $s_0 = 0$. For each $n \geq 1$, the term b_n is chosen according to the value of s_{n-1}:

- If $s_{n-1} \leq L$, then b_n is the first term of (p_k) that has not yet been used.
- If $s_{n-1} > L$, then $b_n = -q$, where q is the first term of (q_k) that has not yet been used.

We refer to these as *positive steps* and *negative steps*, respectively. Under this rule, the p_k are used in the order $p_1, p_2, p_3, \ldots$, and the q_k are used in the order $q_1, q_2, q_3, \ldots$.

Example. Let $L = 1$. Since $s_0 = 0 \leq 1$, the first step is positive and $b_1 = p_1 = 1$. Since $s_1 = 1 \leq 1$, the second step is also positive and $b_2 = p_2 = \frac{1}{3}$. Since $s_2 = \frac{4}{3} > 1$, the third step is negative and $b_3 = -q_1 = -\frac{1}{2}$. The construction continues in this manner.

We emphasize that the rule does not specify the position of each term by an explicit formula. Each choice depends only on whether the current partial sum lies above or below L.

Verification

Claim (A)**.** *The construction contains infinitely many positive steps and infinitely many negative steps.*

Proof. Suppose, for the sake of contradiction, that there are only finitely many negative steps. Then there exists $K \in \mathbb{N}$ such that every step after step K is positive. By the rule, this means $s_n \leq L$ for all $n \geq K$.

On the other hand, for $n > K$, the partial sum s_n is obtained from s_K by adding consecutive terms of (p_k), with no terms subtracted. By Fact 2, these added terms eventually sum to more than $L - s_K$. For such an n, we have $s_n > L$, which is a contradiction.

The argument for positive steps is analogous. If there were only finitely many positive steps, then the partial sums would satisfy $s_n > L$ for all sufficiently large n. However, these partial sums are obtained by subtracting consecutive terms of (q_k), and by Fact 2 they eventually fall to L or below, which is again a contradiction. □

Claim (B)**.** *The series $b_1 + b_2 + b_3 + \cdots$ is a rearrangement of the alternating harmonic series.*

Proof. We must show that every term of the original series appears exactly once.

No term appears more than once. At each step, the rule selects a term that has not yet been used.

Every term appears at least once. Let $k \in \mathbb{N}$. By Claim A, there are infinitely many positive steps, and in particular there is a k-th positive step. Since the positive steps use $p_1, p_2, p_3, \ldots$ in order, the k-th positive step uses p_k. By the same reasoning, the k-th negative step uses q_k. □

Claim (C)**.** *The sequence of partial sums (s_n) converges to L.*

Proof. Let $\varepsilon > 0$. We must find an index beyond which $|s_n - L| < \varepsilon$.

The terms eventually become small. By Fact 1, there exists $K \in \mathbb{N}$ such that $p_k < \varepsilon$ and $q_k < \varepsilon$ for all $k > K$. By Claim A, the construction eventually uses each of $p_1, \ldots, p_K$ and $q_1, \ldots, q_K$; let N be a step by which all of these terms have been used. Every term added after step N is then of the form p_k or $-q_k$ with $k > K$. Consequently,

$$|b_n| < \varepsilon \qquad \text{for all } n > N.$$

The partial sums come within ε of L. By Claim A, there is a positive step after step N, and there is a negative step after that one. Therefore, there exists an index $m > N$ such that step m is positive and step $m+1$ is negative. By the rule,

$$s_{m-1} \leq L \qquad \text{and} \qquad s_m > L.$$

Since $s_m = s_{m-1} + b_m$ and $b_m < \varepsilon$, it follows that

$$L < s_m < L + \varepsilon,$$

and in particular $|s_m - L| < \varepsilon$.

The partial sums remain within ε of L. Suppose that $n \geq m$ and $|s_n - L| < \varepsilon$. We show that $|s_{n+1} - L| < \varepsilon$.

- If $s_n \leq L$, then step $n+1$ is positive, so $0 < b_{n+1} < \varepsilon$. Hence

$$L - \varepsilon < s_n < s_n + b_{n+1} = s_{n+1} \leq L + b_{n+1} < L + \varepsilon.$$

- If $s_n > L$, then step $n+1$ is negative, so $-\varepsilon < b_{n+1} < 0$. Hence

$$L - \varepsilon < L + b_{n+1} < s_n + b_{n+1} = s_{n+1} < s_n < L + \varepsilon.$$

In either case, $|s_{n+1} - L| < \varepsilon$. By induction, $|s_n - L| < \varepsilon$ for every $n \geq m$.

Since $\varepsilon > 0$ was arbitrary, we conclude that $s_n \to L$. □

By Claims B and C, the construction yields a rearrangement of the alternating harmonic series that converges to L. This completes the proof of the theorem. □